

RIO DE JANEIRO GALEAO, Brazil

WMO: 837460

Lat: 22.809S

Lon: 43.244W

Elev: 28

StdP: 14.68

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	58.8	60.6	51.4	56.4	71.4	53.6	61.1	70.9	19.6	75.8	16.7	73.9	3.3	350	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.5	96.9	76.3	94.9	76.5	92.9	76.5	81.9	88.4	80.9	87.5	79.9	86.5	8.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	158.8	85.8	79.1	151.1	84.3	78.6	148.5	83.8	45.7	89.5	44.7	87.6	43.7	86.9	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.2	14.1	DB	54.8	102.1	2.6	2.3	52.9	103.8	51.4	105.1	49.9	106.4	48.0	108.0	(4)
(5)			WB	53.2	83.6	2.5	2.6	51.4	85.5	50.0	87.1	48.6	88.5	46.7	90.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.5	82.1	82.9	80.8	78.2	73.6	71.7	70.9	72.0	73.6	75.8	77.1	80.4	(6)	
	DBStd	5.86	4.38	3.79	3.71	3.88	3.73	3.69	3.99	4.20	4.72	4.80	5.01	4.44	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	10	0	0	0	0	0	2	5	2	1	0	0	0	(9)	
	CDD50	9689	994	920	955	846	731	651	649	682	707	800	813	941	(10)	
	CDD65	4226	529	500	490	396	267	203	189	220	258	335	363	476	(11)	
	CDH74	33343	5431	5308	4468	3025	1310	898	913	1196	1508	2201	2723	4362	(12)	
	CDH80	12118	2308	2278	1602	920	308	206	244	377	521	724	952	1678	(13)	
(14) Wind	WSAvg	6.1	6.4	6.4	6.0	5.7	5.2	4.8	5.1	5.9	6.7	7.1	7.0	6.8	(14)	
(15) Precipitation	PrecAvg	48.3	7.0	5.2	6.1	4.3	2.8	1.8	1.7	1.7	2.7	3.7	5.7	6.2	(15)	
	PrecMax	69.8	14.0	10.9	14.4	11.1	5.1	3.3	4.1	4.9	4.7	10.0	12.9	18.3	(16)	
	PrecMin	18.4	1.5	1.2	1.6	0.7	1.1	0.4	0.2	0.6	0.9	0.8	0.6	1.5	(17)	
	PrecStd	13.5	3.0	2.7	3.8	2.9	1.2	0.7	1.0	1.1	1.2	2.4	3.0	3.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.8	100.2	97.0	94.8	91.0	88.7	89.5	93.0	95.3	95.3	97.0	98.7	(19)	
		MCWB	77.0	76.2	76.6	76.4	73.1	70.8	69.8	69.9	71.4	74.1	75.0	76.8	(20)	
	2%	DB	95.3	96.5	93.4	91.1	86.4	85.6	86.0	88.0	89.9	91.2	93.0	94.8	(21)	
		MCWB	77.3	77.0	77.6	76.5	73.1	70.8	69.7	69.5	71.6	73.9	76.0	77.1	(22)	
	5%	DB	93.1	93.3	90.0	87.8	82.8	82.3	82.6	84.2	85.9	87.4	89.2	91.5	(23)	
		MCWB	77.4	77.4	77.5	76.6	72.4	70.4	69.3	69.5	71.0	73.9	75.4	76.8	(24)	
	10%	DB	90.0	91.0	87.8	85.0	80.5	79.0	78.9	80.5	82.2	84.0	85.6	88.1	(25)	
		MCWB	77.6	77.5	77.7	75.7	71.7	70.1	68.9	69.7	71.0	73.0	75.2	76.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.3	83.9	82.7	81.3	78.3	75.5	74.5	75.6	76.4	79.0	80.4	81.9	(27)	
		MCDB	90.1	93.1	88.8	86.7	83.8	80.7	81.7	82.5	85.3	85.8	88.1	89.1	(28)	
	2%	WB	81.6	81.9	80.9	79.6	76.2	73.7	72.6	73.7	74.6	77.1	78.8	80.1	(29)	
		MCDB	88.2	88.7	87.2	85.1	81.6	79.2	79.1	80.0	83.1	84.6	86.3	87.5	(30)	
	5%	WB	80.5	80.7	79.8	78.3	74.6	72.4	71.2	72.2	73.2	75.7	77.4	79.0	(31)	
		MCDB	87.1	87.5	85.9	83.6	79.7	78.0	77.6	78.4	80.3	82.9	84.5	86.4	(32)	
	10%	WB	79.3	79.5	78.7	77.0	73.1	71.2	70.0	70.7	72.0	74.4	76.1	77.9	(33)	
		MCDB	85.7	86.1	84.6	82.6	78.2	76.8	76.1	77.1	78.9	81.1	82.9	85.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	15.5	13.9	13.8	14.0	15.2	15.8	16.3	14.9	13.6	13.1	13.8	(35)	
		MCDBR	19.3	20.0	18.0	17.7	17.8	20.1	22.4	23.8	23.5	20.5	20.2	19.4	(36)	
	5% WB	MCWBR	6.3	6.4	6.4	6.5	7.3	8.4	9.1	8.8	7.9	7.1	6.9	6.3	(37)	
		MCWBR	15.6	16.5	15.3	14.8	15.1	16.3	17.6	18.2	17.6	16.5	16.3	16.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.414	0.411	0.410	0.398	0.382	0.379	0.381	0.413	0.492	0.470	0.435	0.426	(40)	
		taud	2.465	2.457	2.467	2.487	2.505	2.501	2.476	2.368	2.123	2.247	2.377	2.420	(41)	
	Ebn at Noon		296	292	284	274	265	259	263	265	256	272	288	292	(42)	
		Edh at Noon	38	38	36	32	30	28	30	36	49	46	41	40	(43)	
(44) All-Sky Solar Radiation	RadAvg		1897	1973	1682	1479	1177	1103	1142	1353	1453	1626	1665	1843	(44)	
	RadStd		283	207	141	106	63	81	97	102	184	193	139	171	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	-3.04	N/A	-2.52	-2.99	N/A	N/A	N/A	N/A
	N/A	N/A	-3.04	N/A	-2.52	-2.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page